Topic 7 — Plants, Agriculture and Plant Diseases

Covers syllabus (click to expand)

Plant Taxonomy & Hierarchy (Kingdom to Species, A.P. de Candolle *Vasculares/Cellulares*) | Plant Tissues: Meristematic (Apical, Intercalary, Lateral Cambium) & Permanent (Parenchyma, Collenchyma, Sclerenchyma, Xylem, Phloem) | Transpiration Pull & Ascent of Sap | Morphology & Modifications of Stems (Rhizome, Tuber, Bulb, Corm), Roots (Conical, Fusiform, Napiform, Tuberous) & Leaves | Edible Parts: Clove (unopened floral bud), Saffron (stigma/styles), Okra (capsule), Mango (mesocarp), Coconut (endosperm), Litchi (fleshy aril), Singhara & Cashew (nut seed) | Fruits: True Fruits vs False Fruits (Apple/Pear pome, Cashew peduncle, Jackfruit/Mulberry/Pineapple sorosis, Pepo) | Plant Pigments & Odour Chemistry (Lycopene, Anthocyanin, Caricaxanthin, Curcumin, Capsaicin Scoville scale, Allicin & Propanethial S-oxide in onion/garlic, Fructose) | Economic Plant Fibres (Cotton seed hair vs Sunn/Hemp/Jute bast retting) & Triticale Man-made Cereal | Ecological Plant Types: Xerophytes, Phreatophytes (20–30 m taproots), Hydrophytes (Aerenchyma), Halophytes, Lithophytes, Epiphytes | Photosynthesis Physiology: Light Reaction (Photolysis of Water, Hill), Dark Reaction (Calvin Cycle, RuBisCO), C3 vs C4 (Kranz) vs CAM | Phytohormones: Auxin, Gibberellin (Bolting, Dormancy), Cytokinin, Ethylene, ABA | BNF (Rhizobium, BGA, Azolla) & Insectivorous Plants (Nepenthes khasiana, Drosera, Rafflesia) | Major Crop Diseases: Late Blight, Red Rot, Tikka, Rust, Citrus Canker, Khaira Zinc, Whiptail Mo, Black Heart

Sources baked in: NCERT Class 11 & 12 (Morphology, Anatomy, Plant Physiology, Mineral Nutrition), Lucent General Science (Botany & Agriculture), Ghatnachakra General Studies (Biology Subdivisions & Taxonomy 2025–1990), UPPCS Prelims PYQs (1990–2025), UKPCS, BPSC, RAS, and IAS Prelims **Weight:** ★★★ High — heavily tested: plant taxonomy hierarchy, edible plant parts (clove, saffron, ginger, turmeric, potato vs sweet potato), plant pigments & pungency, true vs false fruits, ecological adaptations (phreatophytes, halophytes), phytohormones, biofertilisers, and crop diseases **Last verified:** September 2026

Current Affairs

Year	Event / Fact	Must-Score Fact
10 February	**World Pulses Day** observed annually on **10 February** by the UN FAO.	Pulses belong to **Fabaceae (Pea family)**; host symbiotic *Rhizobium* in root nodules to fix atmospheric nitrogen and enrich soil.
Discovery	***Musa indandamanensis*** discovered in Little Andaman Island by BSI scientists.	Wild banana attaining **11 metres height** with unique green flowers and orange-coloured fruit pulp thrice the size of regular banana species.
Still Current	FSSAI ban on **Calcium Carbide (CaC_2)** for artificial fruit ripening; promotes safe ripening using **Ethylene gas (C_2H_4)**.	Calcium carbide reacts with moisture to release hazardous **Acetylene gas (C_2H_2)** contaminated with carcinogenic arsenic and phosphorus hydride.
Still Current	**Irish Potato Famine (1845)**: Historical cornerstone of plant pathology caused by **Late Blight of Potato (*Phytophthora infestans*)**, which destroyed Ireland's staple potato crop, leading to mass starvation.	

7.1 Plant Physiology: Photosynthesis, Photolysis & Carbon Assimilation

Master Energetics & Biochemical Reactions

- **Nature of Photosynthesis**: Oxidation-reduction (redox) photochemical reaction in which water is oxidized to release molecular oxygen (O_2) and carbon dioxide is reduced to synthesize carbohydrates (glucose).

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

- **Energy Conversion**: Solar radiation energy is transformed into **chemical potential energy** stored in carbohydrate molecular bonds.
- **Global Photosynthetic Output**: Approximately **90% of Earth's total photosynthesis** is executed by marine and freshwater **algae and phytoplankton** (85% in marine oceans, 5% in freshwater rivers/lakes). Terrestrial forests and grasses contribute the remainder.
- **Source of Evolved Oxygen**: The oxygen gas (O_2) liberated during photosynthesis **originates entirely from the photolysis of water (H_2O)**, proven via isotope tracer experiments (^{18}O), and **not from carbon dioxide**.

Chloroplast Topography & Chlorophyll Chemistry

- **Self-Replication**: Chloroplasts possess their own **extrachromosomal circular DNA and 70S RNA**, conferring autonomous self-replicating capacity (semiautonomous organelle).
- **Chlorophyll Molecule**: Chelated porphyrin ring with a coordinated central **Magnesium (Mg)** ion. Absence of magnesium leads to severe interveinal chlorosis (yellowing). Elements in chlorophyll: C, H, O, N, Mg (**Calcium is NOT present!**).
- **Absorption vs Action Spectrum**:

- Chlorophyll absorbs primarily in the **Blue (440 nm)** and **Red (655 nm)** wavelengths of visible light, while reflecting green light (giving leaves their emerald green appearance).
- **Highest Rate of Photosynthesis**: Occurs in **Red light**; second most effective is **Blue light**; photosynthetic rate is zero in green light.
- **Solarization**: Direct photo-inhibition and photo-oxidation of the photosynthetic apparatus under excessively high light intensities, destroying chlorophyll pigments.
- **Nocturnal Gas Balance**: Photosynthesis ceases in the absence of light, whereas cellular respiration proceeds continuously 24 hours a day. Consequently, trees absorb O_2 and **liberate net CO_2 during the night**; sleeping under dense trees at night leads to local oxygen deprivation.

Light Reactions (Hill Reaction) vs Dark Reactions (Calvin Cycle)

Diagnostic Parameter	Light-Dependent Phase (Hill Reaction)	Light-Independent Phase (Calvin / Blackman Reaction)
Primary Site	**Grana (Thylakoid membranes)** of chloroplast	**Stroma (fluid matrix)** of chloroplast
Light Requirement	Strictly light-dependent	Light-independent (utilizes chemical energy products of light phase)
Key Biochemical Events	Photolysis of water: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$; Electron transport across photosystems I & II; Photophosphorylation	Carbon fixation / carbon assimilation: atmospheric CO_2 reduced into 3-carbon sugars and polymerized into glucose
Key Enzymes	Oxygen-Evolving Complex (OEC containing Mn and Cl), ATP Synthase	**RubisCO** (Ribulose-1,5-bisphosphate carboxylase/oxygenase)
Energetic Currency	Generates **ATP** and **NADPH** across transmembrane proton gradient	Consumes **18 ATP** and **12 NADPH** per single molecule of glucose synthesized

C3, C4 and CAM Photosynthetic Adaptations

- **C3 Pathway (Calvin Cycle)**: Standard pathway in most temperate plants (wheat, rice, barley). Initial stable product is 3-phosphoglycerate (3-PGA, 3 carbon atoms).
- **C4 Pathway (Hatch-Slack Cycle)**: Adapted to high temperature and intense sunlight (sugarcane, maize, sorghum, *Cyperus*). Initial stable carboxylation product is **Oxaloacetic acid (OAA, 4 carbon atoms)**. Possess distinct **Kranz anatomy** (mesophyll and bundle sheath cells). Highly efficient photosynthesizers with near-zero photorespiration.
- **CAM (Crassulacean Acid Metabolism)**: Evolutionary adaptation in desert succulents (Cactus, *Opuntia*, Agave, Pineapple). Stomata remain **open at night** (scotoactive) to absorb CO_2 and synthesize malic acid; stomata remain **tightly sealed during daytime** to prevent transpirational water loss, while internally released CO_2 drives the Calvin cycle.
- **Bacterial Photosynthesis**: Anoxygenic photosynthesis in green sulphur bacteria (*Chlorobium*) utilizing H_2S (hydrogen sulphide) as hydrogen donor instead of H_2O; therefore, **no molecular oxygen is liberated**. Deep-sea sulphur bacteria can utilize invisible infrared radiations for carbon assimilation.

7.2 Plant Nutrition: Essential Macronutrients, Micronutrients & Vascular Transport

Essential Mineral Spectrum (17 Elements)

- **Macronutrients (9 Elements, required in > 10 mmol/kg dry matter)**:
- From Air & Water: Carbon (C), Hydrogen (H), Oxygen (O).
- Primary Soil Nutrients: Nitrogen (N), Phosphorus (P), Potassium (K).
- Secondary Soil Nutrients: Calcium (Ca), Magnesium (Mg), Sulphur (S).
- **Micronutrients / Trace Elements (8 Elements, required in < 10 mmol/kg dry matter)**:
- Iron (Fe), Boron (B), Zinc (Zn), Copper (Cu), Manganese (Mn), Molybdenum (Mo), Chlorine (Cl), Nickel (Ni). (*Sodium Na is NOT essential for most plants!*)

Master Physiological Roles & Deficiency Pathology

Element	Uptake Chemical Form	Physiological Functions	Deficiency Symptoms & Must-Score Pathology
Nitrogen (N)	Nitrate (NO_3^-) predominantly; Ammonium (NH_4^+) in acidic boreal soils	Constituent of amino acids, proteins, chlorophyll, nucleic acids	Severe stunted growth, chlorosis of older leaves
Phosphorus (P)	Dihydrogen phosphate ($H_2PO_4^-$)	Nucleic acids, phospholipids, ATP; **stimulates early root system establishment**, fibrous branching, tillering, and early crop maturity	Dark green or purplish/reddish bronze pigmentation on foliage
Potassium (K)	Potassium ion (K^+)	**Regulates stomatal opening and closing via guard cell turgor pressure**; activates osmotic pumps and enzyme complexes	Marginal leaf burn (tip scorch), chlorosis, weak stems (lodging)

Element	Uptake Chemical Form	Physiological Functions	Deficiency Symptoms & Must-Score Pathology
Sulphur (S)	Sulphate (SO_4^{2-})	Cysteine, methionine, ferredoxin; **crucial for oil synthesis in mustard, rapeseed and groundnut**; legume nodulation	Yellowing of younger leaves, stunted development
Calcium (Ca)	Calcium ion (Ca^{2+})	Component of middle lamella (calcium pectate); membrane stabilization and cell division	**Blossom-end rot** in tomatoes and peppers; death of growing root and shoot tips
Magnesium (Mg)	Magnesium ion (Mg^{2+})	Central coordinating atom of chlorophyll porphyrin ring; activator of RuBisCO and PEPCase	Interveinal chlorosis in older leaves while veins remain green
Iron (Fe)	Ferrous (Fe^{2+}) / Ferric (Fe^{3+})	Electron transport (cytochromes, ferredoxin); essential catalyst for chlorophyll synthesis	Interveinal chlorosis on youngest apical leaves
Boron (B)	Borate (BO_3^{3-} / $B_4O_7^{2-}$)	Pollen tube elongation, cell wall pectin synthesis, calcium translocation, **nodule formation in legumes**	**"Hollow heart"** in beets and peanuts, cracked fruits, failure of flowering
Zinc (Zn)	Zinc ion (Zn^{2+})	Biosynthesis of **Auxin (IAA)**; activator of carbonic anhydrase and alcohol dehydrogenase	**"Khaira disease" in rice (paddy)**; **"Little Leaf"** disease in citrus and mango
Molybdenum (Mo)	Molybdate (MoO_4^{2-})	Core component of **Nitrate Reductase** and **Nitrogenase** enzymes	Whiptail disease in cauliflower; impaired symbiotic nitrogen fixation
Nickel (Ni)	Nickel ion (Ni^{2+})	Structural cofactor of **Urease** enzyme converting urea to ammonia	Necrotic leaf tip lesions due to toxic urea accumulation

Vascular Architecture & Transport Dynamics

- **Stomatal Osmotic Mechanics**: Stomatal pores open when paired **guard cells** absorb water and become turgid (induced by active influx of K^+ ions and malate). At night, K^+ and water efflux cause guard cells to become flaccid, sealing the stomatal aperture.
- **Transpiration Pull**: Evaporative water loss through stomata creates a severe negative pressure (suction pull) in the xylem conduits. Supported by water's cohesion, adhesion, and surface tension (**Cohesion-Tension Theory of Dixon and Joly**), this pull lifts continuous water columns to heights exceeding 100 metres in tall redwood trees.
- **Xylem (Water Conduction)**:
- Formed of dead lignified cells (tracheids, vessel elements, xylem fibres) with living xylem parenchyma.
- Transports water and dissolved soil mineral nutrients **unidirectionally upward** (roots → stems → leaves).
- Permanent internal trunk tissue: A metal nail hammered 1 metre above ground into a tree trunk **remains at exactly the same 1-metre height after years** because vertical elongation occurs strictly at apical meristems, while trunk girth expands via lateral vascular cambium outward around the nail.
- **Phloem / Bast (Organic Nutrient Conduction)**:
- Composed of living cells (**sieve tubes, companion cells**, phloem parenchyma) and dead sclerenchymatous phloem fibres.
- Transports photosynthetic assimilates (**sucrose**) **bidirectionally** from source (photosynthetic leaves) to sink (roots, growing fruits, storage tubers).
- **Girdling / Ringing Experiment**: Removing a complete ring of bark (containing phloem) down to the vascular cambium around a tree trunk cuts off food supply to the roots. While water continues ascending through inner xylem, **the roots starve of metabolic carbohydrates**, causing the tree to dry up and die.
- **Parasitic Angiosperms (*Cuscuta* / Amarbel)**:
- Non-green, leafless, rootless, yellow twining stem parasite (Family Convolvulaceae).
- Develops specialized microscopic adventitious root-like suckers called **haustoria** that penetrate host stems, bridging into host **xylem (for water/minerals)** and **phloem (for organic nutrients)**.
- **Crop Logging**: Diagnostic agronomic method founded by H.F. Clements for monitoring sugarcane nutritional status in Hawaii, tracking tissue nutrient concentrations to optimize fertilizer applications throughout the vegetative cycle.

7.3 Phytohormones: Growth Promoters, Inhibitors & Fruit Ripening Agents

Comparative Spectrum: Endogenous Plant Growth Regulators

Hormone Class	Chemical Representative	Primary Sites of Synthesis	Physiological Functions & Agronomic Applications
Auxins	**Indole-3-Acetic Acid (IAA)** (Natural); **NAA, IBA, 2,4-D, 2,4,5-T** (Synthetic)	Shoot apical buds, young expanding leaves	- **Apical Dominance**: Promotes terminal bud growth while suppressing axillary buds. - **Phototropism & Geotropism**: Cell elongation on shaded side. - **Root Initiation**: Indole-3-butyric acid (IBA) stimulates rapid adventitious rooting in stem cuttings. - **Parthenocarpy**: Induces seedless fruit formation in tomatoes. - **Weedicide**: Synthetic **2,4-D** selectively eradicates broadleaf (dicot) weeds in cereal fields.

Hormone Class	Chemical Representative	Primary Sites of Synthesis	Physiological Functions & Agronomic Applications
Gibberellins	**Gibberellic Acid (GA_3)**	Plastids of young leaves, developing seeds, roots	- **Internodal Stem Elongation**: Overcomes dwarfism; sprayed on sugarcane to dramatically lengthen internodes and boost cane yield. - **Bolting**: Induces internode lengthening prior to flowering in rosette plants (cabbage, beet). - **Breaks Seed & Bud Dormancy**: Stimulates α-amylase synthesis in barley aleurone layer. - Discovered from the fungus *Gibberella fujikuroi* causing "Bakanae" (foolish seedling) disease in rice.
Cytokinins	**Kinetin, Zeatin**	Root apical meristems, endosperm of seeds	- **Cell Division (Cytokinesis)**: Promotes mitosis and shoot organogenesis. - **Delays Senescence (Richmond-Lang Effect)**: Retards chlorophyll breakdown and protein degradation; preserves freshness in cut flowers and vegetables. - Abundantly present in **coconut milk / liquid endosperm**.
Abscisic Acid (ABA)	Sesquiterpene carboxylic acid	Chloroplasts of leaves, vascular tissues	- **"Stress Hormone"**: Triggers rapid stomatal closure under drought stress (**antitranspirant**). - **Induces Dormancy**: Maintains seed and winter bud dormancy; acts as a physiological **anti-gibberellin**. - Promotes leaf senescence and fruit abscission.
Ethylene	**Gaseous hydrocarbon (C_2H_4)**	Ripening fruit tissues, senescent foliage, nodes	- **Fruit Ripening**: Triggers the **"respiratory climacteric"** (surge in cellular respiration) in bananas, apples, mangoes. - Promotes leaf, flower, and fruit abscission. - Stimulates latex exudation in rubber trees (*Hevea brasiliensis*). - Synthesized biochemically from the amino acid **methionine**.

Commercial Fruit Ripening: Ethylene vs Acetylene

- **Ethrel / Ethephon (2-Chloroethylphosphonic Acid)**: Water-soluble liquid chemical widely approved for commercial fruit ripening. Once absorbed into plant tissues, it breaks down to release **natural ethylene gas** uniformly and safely without toxic residues.
- **Calcium Carbide (CaC_2)**: Cheap industrial agent that reacts with moisture to liberate **Acetylene gas (C_2H_2)**:

$$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2 \uparrow$$

- Acetylene acts as an analog to ethylene but is far less potent and **strictly prohibited under the Food Safety and Standards Act (FSSAI)** because commercial calcium carbide contains hazardous traces of **Arsenic hydride (AsH_3)** and **Phosphorus hydride (PH_3)**, posing severe neurological and carcinogenic hazards.

7.4 Plant Reproduction: Vegetative Propagation, Floral Architecture & Embryogenesis

Asexual Reproduction & Vegetative Techniques

- **Clonal Fidelity**: Vegetative propagation produces offspring that are exact genetic copies (**clones**) of the parent plant, preserving elite horticultural characteristics across generations.
- **Stem Cuttings**: Portions of stem with nodal buds inserted in soil to develop adventitious roots; practiced in **Sugarcane, Grapes, Rose, Bougainvillea, Carnation, Cocoa**.
- **Layering / Air Layering (Gootee / Marcottage)**: Encouraging adventitious root development on an aerial stem while still attached to the parent plant by girdling bark and wrapping in moist sphagnum moss; widely used in **Jasmine, Litchi, Pomegranate, Strawberry**.
- **Grafting**: Joining tissues of two distinct plants so their vascular systems fuse (**inosculation**):
- **Scion**: Upper shoot system selected for desirable flowers or fruits.
- **Rootstock**: Lower rooted stem selected for disease resistance and vigorous root anchorage.
- Common in **Sweet Orange, Apple, Mango, Walnut**.
- **Apomixis**: Asexual seed formation bypassing meiosis and syngamy; produces seeds genetically identical to maternal parent, enabling cost-effective commercial propagation of hybrid crops without segregation of hybrid vigor.

Floral Morphological Architecture

- **Flower**: Specialized reproductive shoot mediating the sexual life cycle of angiosperms.
- **Four Floral Whorls**: 1. **Calyx (Sepals)**: Outermost whorl, protects floral bud; when colored, termed *petaloid*. 2. **Corolla (Petals)**: Second whorl, vividly colored to attract pollinating insects and birds. 3. **Androecium (Male)**: Composed of stamens (slender filament topped by a bilobed **anther**). Anthers contain microspore mother cells dividing by meiosis to produce haploid pollen grains; **anthers are the finest botanical tissue to observe plant meiosis under microscope**. 4. **Gynoecium / Pistil (Female)**: Composed of carpels comprising a landing **stigma**, slender **style**, and basal swollen **ovary** containing ovules.
- **Dichogamy**: Temporal separation of anther dehiscence and stigma receptivity within a bisexual flower to prevent self-pollination:
- **Protandry**: Anthers mature and discharge pollen before stigma becomes receptive (e.g. Sunflower, Cotton).
- **Protogyny**: Stigma matures and is fertilized before anthers shed pollen (e.g. *Ficus*, Magnolia).

Pollination Ecology & Vectors

- **Self-Pollination**: Autogamy (within same flower) and **Geitonogamy** (transfer of pollen from anther of one flower to stigma of another flower on the **same plant**; genetically equivalent to self-pollination).
- **Cross-Pollination (Xenogamy)**: Transfer of pollen to stigma of a flower on a genetically distinct plant; enhances genetic diversity.
- **Pollinating Vectors**:
- **Entomophily (Insects)**: Bees (**world's primary pollinator**), butterflies, moths; flowers display fragrant scents, nectar, and vibrant yellow/blue colors.
- **Ornithophily (Birds)**: Hummingbirds, sunbirds; flowers are odorless, bright red/orange/yellow with tubular corollas.
- **Chiropterophily (Bats)**: Nocturnal large, dull-colored, heavily scented flowers with copious nectar.
- **Anemophily (Wind)**: Grasses, maize; light, non-sticky, smooth pollen produced in huge abundance; feathery stigmas.

Angiosperm Double Fertilization & Seed Genesis

- **Double Fertilization (Discovered by S. Nawaschin)**:
- Pollen tube delivers two haploid sperm nuclei into the ovule embryo sac:

$$\text{Sperm 1}(n) + \text{Egg } (n) \xrightarrow{\text{Syngamy}} \text{Zygote } (2n) \longrightarrow \text{Embryo}$$

$$\text{Sperm 2}(n) + \text{Central Cell Polar Nuclei } (2n) \xrightarrow{\text{Triple Fusion}} \text{Endosperm } (3n)$$

- **Endosperm Ploidy**: Triploid ($3n$); provides essential nutrition to the developing embryo.
- In Maize (*Zea mays*): $2n = 20$, gamete $n = 10$, endosperm $3n = 30$ **chromosomes**.
- Hexaploid Wheat (*Triticum aestivum*): $2n = 6x = 42$ chromosomes possessing the **ABD genome** (AABBDD).
- **Post-Fertilization Fate**:
- **Ovary $\longrightarrow$ Fruit**
- **Ovule $\longrightarrow$ Seed**
- **Integuments $\longrightarrow$ Seed coat (Testa/Tegmen)**
- **Radicle $\longrightarrow$ Primary root**
- **Plumule $\longrightarrow$ Embryonic shoot** (NOT root tip!).
- **Seed Multiplication Hierarchy (Certification Chain)**:

$$\textbf{Nucleus Seed} \longrightarrow \textbf{Breeder Seed (Golden Yellow Tag)} \longrightarrow \textbf{Foundation Seed (White Tag)} \longrightarrow \textbf{Certified Seed (Azure Blue Tag)}$$

- **Censer Mechanism**: Wind swaying dry capsule with apical pores shaking out minute seeds (characteristic of **Poppy / *Papaver somniferum***).
- **Seed Germination Requirements**: Strictly requires **Moisture/Water, Favourable Temperature, and Oxygen**; light and soil are non-essential for initial seed germination.

7.5 Economic Botany: Cash Crops, Biofuels, Petro-Crops, Medicinal Plants & Plant Products

Commercial Crop Cultivars & Agricultural Geography

- **Amla (Indian Gooseberry, *Phyllanthus emblica*)**: Richest botanical source of Vitamin C; improved varieties: **Kanchan, Krishna, Banarasi, Chakaiya, NA-7**.
- **Mango (*Mangifera indica*)**:
- **Ratna**: Hybrid variety developed from cross of **Neelum** × **Alphonso**.
- **Amrapali**: Hybrid developed from cross of **Dasheri** × **Neelum**.
- **Okra / Bhindi (*Abelmoschus esculentus*)**: **Kashi Lalima** (red-hued variety bred by IIVR Varanasi, rich in anthocyanins, iron, and calcium).
- **Sugarcane (*Saccharum officinarum*)**: **Co. 1148** (prominent drought-tolerant, high-tillering variety of eastern Uttar Pradesh).
- **Macaroni / Durum Wheat**: **Triticum durum** (tetraploid wheat, $2n = 4x = 28$, ideal for semolina, pasta, and macaroni).

Biofuels, Petro-Crops & National Biofuel Policy

- **Petro-Crops**: Plants yielding biomass rich in combustible **liquid hydrocarbons and lipids**:
- **Jatropha (*Jatropha curcas*)**: Non-edible drought-hardy shrub; seeds yield up to 35–40% oil converted to biodiesel via transesterification.
- **Karanja (*Pongamia pinnata*)**: Leguminous tree; red seed oil used for biodiesel and dermal remedies.
- **Neem (*Azadirachta indica*)**: Seed oil used for biodiesel and biopesticides.
- **National Policy on Biofuels 2018 (India)**:
- Permitted Feedstocks for 1G Ethanol: Sugarcane juice, B-heavy molasses, sugar beet, sweet sorghum, corn, cassava, rotten potatoes, damaged food grains (wheat, broken rice). (*Groundnut seeds and horsegram are excluded!*)
- 4th Generation Biofuels: Synthesized using **genetically modified photosynthetic microorganisms**, notably **GM Algae** and cyanobacteria.
- India's First Commercial Biodiesel Plant: Established at **Kakinada, Andhra Pradesh**.
- Ethanol Geopolitics: United States is world's #1 ethanol producer (principal feedstock: **Corn/Maize**); Brazil is #2 (principal feedstock: **Sugarcane**).

- **Canola Oil**: Special low-acid rapeseed cultivar bred in Canada; defined by **low Erucic acid content (<2%)** and low glucosinolates, making it exceptionally heart-healthy.

Commercial Timber & Gymnosperm Geography

- **Himalayan Timber Conifers**: **Pine (*Pinus*), Fir (*Abies*), Spruce (*Picea*), Deodar / Cedar (*Cedrus deodara*), Cypress, Larch**; all belong to the division **Gymnosperms** (bearing naked seeds unenclosed by fruit ovaries).
- **Chilgoza (Neozo)**: Edible seed rich in proteins, fats, and carbohydrates harvested from the cones of the **Chilgoza Pine (*Pinus gerardiana*)**, native to northwestern Himalayas.
- **Red Sanders (*Pterocarpus santalinus*)**: Endemic endangered leguminous hardwood tree restricted to the **Eastern Ghats (Seshachalam Hills of Andhra Pradesh)**; valued for red timber and santalin dye.

Medicinal Flora & Active Pharmacological Principles

- **Neem (*Azadirachta indica*)**: "The Village Pharmacy"; contains **Azadirachtin** (potent antifeedant, insecticidal and hepatoprotective principle) and **Nimbolide** (anti-malarial and cytotoxic terpenoid lactone).
- **Tulsi (*Ocimum tenuiflorum / O. sanctum*)**: Family Lamiaceae; medicinal value derived from high concentrations of **phenolic compounds and flavonoids** (eugenol, carvacrol, apigenin, luteolin).
- **Turmeric (*Curcuma longa*)**: Active polyphenol **Curcumin**; powerful anti-inflammatory, anti-arthritic, antioxidant, suppresses Alzheimer's amyloid accumulation.
- **Brahmi (*Bacopa monnieri*)**: Renowned Ayurvedic **nootropic / brain tonic**; enhances synaptic transmission, memory, and learning speed; used in ADHD and anxiety.
- **Sadabahar (*Catharanthus roseus / Vinca rosea*)**: Source of chemotherapeutic indole alkaloids **Vincristine and Vinblastine** (used in treating Hodgkin's disease and acute lymphocytic leukemia); traditional oral hypoglycemic.
- **Mint / Pudina (*Mentha arvensis*)**: Richest source of **Menthol** (monoterpene alcohol); carminative, antispasmodic, stimulates bile secretion.
- **Medicinal Mushrooms**: Source of bioactive polysaccharides (β-glucans); **Psilocybin mushrooms** produce psychoactive/psychedelic alkaloids (psilocybin, psilocin); species like *Panellus stipticus* and *Omphalotus olearius* display **bioluminescence** via luciferin-luciferase oxidation.

Plant Narcotics, Stimulants & Commercial Derivatives

- **Opium Poppy (*Papaver somniferum*)**:
- **Opium**: Dried latex harvested by incising the **unripe fruit capsules** of the poppy plant.
- Contains ~12% **Morphine** (potent narcotic analgesic, central nervous system depressant); chemically converted into **Heroin** (diacetylmorphine); other alkaloids include **Codeine, Papaverine, Noscapine**.
- **Cinchona Tree (*Cinchona officinalis*, Family Rubiaceae)**:
- **Quinine**: Antimalarial alkaloid drug extracted from the **stem bark** of the tree; historically crucial against *Plasmodium*; largely replaced by synthetic **Chloroquine**.
- **Arteether**: Potent antimalarial sesquiterpene derivative synthesized from artemisinin, isolated from **Artemisia annua**.
- **Caffeine**: Purine alkaloid central nervous system stimulant found naturally in **both tea leaves (*Camellia sinensis*) and coffee beans (*Coffea arabica*)**, as well as cacao beans and added to cola soft drinks.
- **Chicory (*Cichorium intybus*)**: Herbaceous plant whose roasted, ground **roots** are blended into coffee to diminish bitterness and impart rich aroma.
- **Kuttu Flour (Buckwheat)**: Ground from seeds of *Fagopyrum esculentum* (Family Polygonaceae); rich in protein and starch; an edible pseudocereal (unrelated to true Poaceae grasses).
- **Saffron (*Kesar*)**: World's costliest spice; consists exclusively of the dried crimson **stigmas** and styles of the autumn crocus flower (*Crocus sativus*).
- **Asafoetida (*Hing*)**: Dried oleo-gum resin harvested by tapping the **rhizomes and fleshy tap roots** of *Ferula foetida* and related species (Family Apiaceae).
- **Cork**: Harvested sustainably from the thick dead suberized outer bark of the oak tree **Quercus suber** (native to western Mediterranean).
- **Cotton**: Staple fiber composed of **91% pure cellulose** originating as epidermal seed hairs inside the fruit bolls of *Gossypium* (Family Malvaceae).
- **Laminaria (Brown Kelp)**: Marine brown seaweed harvested as the foremost natural biological source of **Iodine**.

7.6 Plant Pathology & Crop Protection: Viral, Bacterial, Fungal & Abiotic Diseases

Dissemination Vectors & Plant Pathology Foundations

- **Phytopathology**: The scientific study of plant diseases caused by biotic pathogens (fungi, bacteria, viruses, nematodes, phytoplasmas) and abiotic physiological stresses (mineral deficiencies, temperature extremes, toxic gases).
- **Disease Transmission Vectors**: Plant pathogens disseminate dynamically via:
- **Atmospheric Vectors**: Wind currents, dust blowing, rain splashing.
- **Biological Vectors**: Insects (leafhoppers, aphids, whiteflies), birds, nematodes, and human agricultural operations (grafting, pruning tools, contaminated farm machinery).
- **Edaphic & Propagative Vectors**: Soil-borne resting spores, infected seed lots, contaminated irrigation run-off.

- **Universal Host Susceptibility**: Viruses infect all forms of life, including higher angiosperms, gymnosperms, fungi (**mycoviruses**), and bacteria (**bacteriophages**).

Master Reference Table: Biotic Plant Diseases & Causative Pathogens

Crop Host	Disease Common Name	Causative Pathogen / Class	Vector / Transmission Mode	Landmark Pathology & Agronomic Symptoms
Wheat	**Black Rust (Stem Rust)**	*Puccinia graminis tritici* (Fungus)	Wind-borne urediniospores; heteroecious (alternate host: Barberry)	Reddish-brown to black elongated pustules on stems and leaf sheaths; shrivelled grain.
Wheat	**Loose Smut**	*Ustilago nuda tritici* (Fungus)	Internally seed-borne	Floral earheads transformed into loose black powdery masses of teliospores.
Wheat	**Karnal Bunt (Partial Bunt)**	*Tilletia indica* (Smut Fungus)	Air-borne and soil-borne sporidia; first found in Karnal (1931)	Affects grains partially; crushed bunted grains emit a foul **fishy odor (Trimethylamine)**; ruins flour palatability.
Wheat	**Ear Cockle / Sehun (Sahu)**	*Anguina tritici* (**Nematode**)	Seed-borne galls / soil	Grains converted into hard, dark-brown or black cockle galls; distorted leaves.
Paddy (Rice)	**Rice Blast**	*Magnaporthe oryzae* (*Pyricularia oryzae*) (Fungus)	Airborne conidia; high humidity and nitrogen	Diamond/spindle-shaped lesions with ash-grey centers on leaves, nodal rot, and "neck blast" snapping panicles.
Paddy (Rice)	**Brown Leaf Spot**	*Sphaerulina oryzina* (*Helminthosporium oryzae*) (Fungus)	Airborne spores; responsible for **Bengal Famine of 1943**	Oval, brown spots with yellow halos across foliage; severe grain discoloration.
Paddy (Rice)	**False Smut**	*Ustilaginoidea virens* (Fungus)	Spores infect floral organs	Individual grains transformed into large velvety greenish-orange to black fungal masses.
Paddy (Rice)	**Tungro Disease**	Dual Viral Complex: **RTSV** (Spherical) + **RTBV** (Bacilliform)	**Green Leafhopper (*Nephotettix virescens*)**	Severe stunting, orange-yellow leaf discoloration from tip downward, sterile panicles.
Paddy (Rice)	**Khaira Disease**	**Abiotic: Zinc (Zn) Deficiency**	Physiological soil deficiency	Rusty brown pigmentation on older leaves, stunted tillering; treated via Zinc Sulphate spray.
Sugarcane	**Red Rot**	*Colletotrichum falcatum* (Fungus)	Infected seed setts, irrigation water	Internal stem tissues turn blood red with discontinuous white patches; emits a **sour alcoholic fermentation odor**.
Sugarcane	**Sugarcane Leafhopper Pest**	*Pyrilla perpusilla* (**Insect Pest**)	Gregarious nymphs and adults on leaf undersides	Sucks phloem sap, reducing sucrose by up to 30%; excretes sticky honeydew attracting **sooty mould** fungus.
Potato	**Late Blight**	*Phytophthora infestans* (Oomycete Fungus)	Air-borne sporangia; caused **Irish Potato Famine (1845)**	Water-soaked dark necrotic leaf lesions, rotting tubers with dry/wet brown rot.
Potato	**Early Blight**	*Alternaria solani* (Fungus)	Airborne conidia	Concentric target-board concentric rings on leaves; defoliation.
Potato	**Bacterial Wilt / Ring Rot**	*Pseudomonas solanacearum* (Bacterium)	Soil-borne and tuber-borne	Vascular browning of xylem ring; milky bacterial ooze upon squeezing stem.
Potato	**Black Heart (Krishnakant)**	**Abiotic: Acute Oxygen Deficit**	Inadequate ventilation in cold storages or high field heat	Core tissues undergo internal suffocation, dying from inside out and turning jet black.
Mustard	**White Rust**	*Albugo candida* (Oomycete Fungus)	Soil-borne oospores, airborne sporangia	Prominent white, blister-like pustules on leaf undersides, stems, and hypertrophied floral heads.
Groundnut	**Tikka Disease (Leaf Spot)**	*Cercospora personata* & *C. arachidicola* (Fungi)	Wind-blown conidia and infected plant debris	Circular dark necrotic spots with distinct chlorotic yellow halos; extensive defoliation.
Bajra (Pearl Millet)	**Green Ear Disease (Downy Mildew)**	*Sclerospora graminicola* (Fungus)	Soil-borne oospores; obligate parasite	Floral earheads transform into a mass of **twisted green leafy structures** resembling a broom.
Bajra (Pearl Millet)	**Ergot Disease**	*Claviceps microcephala* (Fungus)	Airborne conidia during flowering	Sticky honey-dew exudes from spikelets, replaced by dark sclerotial bodies containing toxic ergot alkaloids.
Okra (Bhindi)	**Yellow Vein Mosaic**	Bhendi Yellow Vein Mosaic Virus (BYVMV)	**Whitefly (*Bemisia tabaci*)**	Vein clearing followed by intense complete yellow network of veins; small chlorotic yellow fruits.
Citrus	**Citrus Canker**	*Xanthomonas axonopodis* pv. *citri* (Bacterium)	Rain splash, wind, citrus leafminer	Raised, corky, crater-like necrotic lesions on leaves, twigs, and rind with yellow halo.

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Pigeon Pea (Arhar)	**Wilt Disease (Ukhtha)**	*Fusarium oxysporum* f. sp. *udum* (Fungus)	Soil-borne vascular pathogen	Black streak in xylem conduits; vascular blockage causes rapid foliage drooping and death.
Crucifers (Cabbage)	**Black Rot**	*Xanthomonas campestris* (Bacterium)	Hydathodes entry via rain splash	V-shaped chlorotic to necrotic lesions on leaf margins; vascular blackening.
Mango (Alphonso)	**Spongy Tissue**	**Physiological Ripening Breakdown**	Peculiar to coastal Maharashtra/South India	Inactivation of ripening enzymes; internal mesocarp turns white, spongy, acidic and corky.

Diagnostic Deficiencies & Abiotic Crop Disorders

- **Zinc (Zn) Deficiency**:
- *Khaira Disease of Paddy*: Punctate rusty reddish-brown patches across leaf blades.
- *White Bud of Maize*: Apical unfolding leaves emerge bleached yellowish-white due to failure of chlorophyll and auxin synthesis.
- *Little Leaf of Cotton & Citrus*: Extreme reduction in leaf lamina size with shortening of internodes.
- **Boron (B) Deficiency**:
- *Internal Cork / Browning in Apple*: Hard corky lesions inside fruit flesh.
- *Browning in Cauliflower*: Water-soaked internal pith browning.
- *Heart Rot of Sugar Beet*: Death and blackening of young central crown leaves.
- **Molybdenum (Mo) Deficiency**:
- *Whiptail of Cauliflower & Cabbage*: Lamina fails to develop, leaving only strap-like bare midribs.
- **Oxygen (O_2) Deficiency**:
- *Black Heart of Potato*: Sub-atmospheric oxygen in deep storage bins halts cellular respiration, killing internal pith cells.

Bilingual Terminology -- Plants, Agriculture and Diseases

English Term	Hindi Term	Key Anchor
Photosynthesis (प्रकाश संश्लेषण)	प्रकाश-संश्लेषण	Light energy -> chemical energy; CO2 + H2O -> glucose + O2
Chlorophyll (पर्णहरिम)	क्लोरोफिल	Green pigment in plants; absorbs red and blue light
Transpiration (वाष्पोत्सर्जन)	वाष्पोत्सर्जन	Evaporation of water from stomata; cooling + mineral transport
Stomata (रन्ध्र)	रन्ध्र	Pores on leaf surface; controlled by guard cells; CO2 in, O2 out
Root (जड़)	जड़ / मूल	Anchors plant; absorbs water and minerals; some store food
Xylem (जाइलम / दारु)	जाइलम	Conducts water and minerals from roots to leaves (upward)
Phloem (फ्लोएम / पोषवाह)	फ्लोएम	Conducts food (sucrose) from leaves to all parts (downward + upward)
Pollination (परागण)	परागण	Transfer of pollen from anther to stigma
Fertilisation in Plants (पुष्प निषेचन)	निषेचन	Fusion of pollen nucleus with egg nucleus in ovule
Seed (बीज)	बीज	Ripened ovule; contains embryo + endosperm
Fruit (फल)	फल	Ripened ovary; protects seeds; aids dispersal
Nitrogen Fixation (नाइट्रोजन स्थिरीकरण)	नाइट्रोजन स्थिरीकरण	Atmospheric N2 converted to NH3/NH4+; by bacteria/lightning
Legume (फलीदार पौधा)	फली वाला पौधा	Plant with root nodules containing Rhizobium bacteria (N-fixation)
Rhizobium (राइजोबियम)	राइजोबियम	Nitrogen-fixing bacteria in root nodules of legumes
Mycorrhiza (माइकोराइज़ा)	कवक-मूल	Symbiotic fungi-root association; improves phosphate absorption
Plant Hormone (पादप हार्मोन)	पादप हार्मोन	Chemical regulator in plants; auxin, gibberellin, cytokinin, etc.
Auxin (ऑक्सिन)	ऑक्सिन	Plant hormone; promotes cell elongation; phototropism
HYV Seeds (उच्च उपज वाली किस्में)	उच्च उपज किस्में	High Yielding Variety seeds; basis of Green Revolution
Green Revolution (हरित क्रांति)	हरित क्रान्ति	1960s; HYV + fertiliser + irrigation; M.S. Swaminathan (India)

One-Liner Revision -- Plants, Agriculture and Diseases

#	Fast Fact
1	Photosynthesis equation: 6CO2 + 6H2O + light -> C6H12O6 + 6O2
2	Chlorophyll absorbs **red** and **blue-violet** light; reflects **green** (hence green colour)
3	Light reactions occur in **thylakoid** membranes (grana of chloroplast)
4	Dark reactions (Calvin cycle) occur in **stroma** of chloroplast
5	Transpiration = evaporation of water through stomata; creates suction for water uptake
6	Xylem = water + minerals transport (root to leaves); dead cells; one-way
7	Phloem = food (sucrose) transport; living cells; bidirectional
8	Rhizobium = nitrogen-fixing bacteria in legume root nodules; mutualistic
9	Azotobacter = free-living nitrogen-fixing bacteria in soil
10	Cyanobacteria (blue-green algae) = nitrogen fixers; e.g., Anabaena in rice fields
11	Auxin = phototropism (bending towards light); produced at shoot tip
12	Gibberellin = promotes stem elongation; used to break seed dormancy
13	Ethylene = fruit ripening gas; only gaseous plant hormone
14	Abscisic acid = stress hormone ("stress hormone of plant"); causes wilting + dormancy
15	Green Revolution: M.S. Swaminathan (India) + Norman Borlaug (Mexico; Nobel 1970)
16	HYV wheat = Sonalika, Kalyan Sona; HYV rice = IR8 ("Miracle Rice")
17	Vermiculture = use of earthworms to decompose organic waste (vermicompost)
18	Algal bloom = excessive algae growth due to eutrophication (excess nitrate/phosphate)
19	Eutrophication = excess nutrients in water -> algae bloom -> O2 depletion -> fish death
20	Most common plant disease agent = **Fungi** (then bacteria, then viruses)
21	Rust of wheat = *Puccinia graminis* (fungus)
22	Blast of rice = *Pyricularia oryzae* (fungus) -- most destructive rice disease
23	Smut of wheat = *Ustilago tritici* (fungus)
24	Citrus canker = *Xanthomonas citri* (bacteria)
25	Tobacco Mosaic Virus (TMV) = first virus discovered (Dmitri Ivanovsky, 1892)
26	Potato virus = causes "late blight of potato" (*Phytophthora infestans*) = Irish Famine
27	Hardwood (dicot wood) = Teak, Sal, Rosewood; Softwood (conifer) = Pine, Spruce, Deodar
28	Bryophytes = "amphibians of plant kingdom" (e.g., moss, liverwort); no vascular tissue
29	Gymnosperms = naked seeds (no fruit); e.g., Pine, Cycas, Ginkgo
30	Angiosperms = enclosed seeds in fruit; most advanced; ~2.5 lakh species